

BRENHAM, TX, USA

WMO: 722553

Lat: 30.219N

Lon: 96.374W

Elev: 308

StdP: 14.53

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.0	32.0	16.6	12.9	39.9	20.6	15.6	43.8	25.0	55.4	22.3	55.2	4.7	350	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.4	99.2	74.6	97.3	75.2	95.4	75.8	80.2	88.3	79.3	87.6	78.5	86.9	7.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	151.3	81.9	77.6	145.2	81.2	76.9	141.6	81.1	43.9	87.9	43.0	88.3	42.3	86.9	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.2	17.8	15.9	DB	23.2	102.4	4.5	2.7	20.0	104.3	17.3	105.9	14.8	107.3	11.6	109.3
			WB	22.0	81.2	4.6	1.3	18.6	82.1	15.9	82.8	13.3	83.6	10.0	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.6	51.9	55.7	62.6	68.5	76.2	82.7	84.3	85.2	80.2	71.2	61.5	54.1
	DBStd	13.86	9.93	10.39	8.55	6.90	5.64	3.32	3.21	3.23	4.69	7.74	9.36	9.67
	HDD50	245	95	56	11	1	0	0	0	0	0	0	16	66
	HDD65	1468	420	287	147	43	3	0	0	0	0	33	173	362
	CDD50	7400	155	217	403	556	813	980	1065	1093	906	659	361	192
	CDD65	3147	14	28	74	148	352	530	600	628	456	226	68	23
	CDH74	33320	60	129	404	1085	3296	6115	7152	7707	4699	2081	475	117
	CDH80	15479	3	14	57	230	1255	3054	3683	4167	2193	729	86	8
(14) Wind	WSAvg	6.0	6.7	7.3	7.2	7.5	6.9	5.6	5.1	4.7	4.1	4.9	5.8	6.4
(15) Precipitation	PrecAvg	43.6	3.1	3.1	2.8	3.3	4.8	4.7	2.2	2.8	4.8	4.4	4.2	3.3
	PrecMax	64.6	7.9	8.2	8.7	8.3	13.1	15.7	7.8	11.2	15.6	17.5	10.6	11.2
	PrecMin	21.3	0.2	0.2	0.2	0.2	0.1	0.0	0.1	0.0	0.2	0.4	0.6	0.6
	PrecStd	10.6	2.0	1.9	1.9	2.1	3.0	3.4	1.9	2.4	3.3	4.0	2.7	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.3	82.2	84.4	88.3	94.6	99.8	100.2	101.7	99.2	93.5	86.0	81.3
		MCWB	65.9	67.0	67.7	70.8	74.4	73.7	74.2	74.7	72.1	74.4	72.3	68.8
	2%	DB	75.2	78.0	81.5	85.0	91.2	97.1	98.5	99.6	96.3	90.4	82.2	77.2
		MCWB	63.8	66.3	67.2	70.0	74.2	73.9	75.0	75.1	74.1	74.4	70.0	67.8
	5%	DB	72.1	74.7	78.9	82.3	89.8	94.7	96.6	97.7	93.2	87.6	79.4	73.9
		MCWB	63.7	65.3	66.4	69.0	73.8	74.8	75.7	76.0	74.7	73.5	67.6	67.3
	10%	DB	69.2	72.0	75.4	80.7	87.5	92.4	94.6	96.0	90.8	84.0	76.5	71.8
		MCWB	62.2	64.6	65.5	67.9	73.2	75.2	75.9	76.2	74.3	70.6	67.7	65.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.2	71.3	72.4	76.3	79.2	80.7	81.2	81.3	80.4	78.8	75.7	73.2
		MCDB	73.9	75.9	77.5	82.2	85.7	87.9	90.1	89.5	86.3	86.0	80.8	77.0
	2%	WB	68.1	69.8	71.2	74.1	77.7	79.4	79.9	80.0	79.0	77.4	73.8	71.1
		MCDB	72.6	74.1	76.4	80.4	85.0	86.8	88.5	89.0	85.8	83.9	78.7	75.2
	5%	WB	66.3	68.2	69.8	72.7	76.4	78.5	79.0	79.1	78.1	76.3	71.8	69.0
		MCDB	70.7	72.9	74.5	78.8	83.9	86.1	87.6	88.4	85.5	82.9	76.8	72.9
	10%	WB	63.4	66.2	68.3	71.3	75.4	77.7	78.3	78.3	77.2	74.9	69.9	66.1
		MCDB	66.1	70.7	73.5	76.9	82.8	85.7	86.9	87.7	84.9	81.4	74.2	70.5
(35) Mean Daily Temperature Range	5% DB	MDBR	20.1	19.3	20.0	20.4	18.8	19.1	18.7	20.4	19.8	21.9	21.0	20.9
		MCDBR	23.0	21.3	20.8	20.7	20.1	22.0	21.6	22.6	22.1	21.6	22.4	22.6
		MCWBR	13.8	12.3	10.0	9.4	6.5	5.3	5.4	5.4	6.1	8.9	12.3	15.4
	5% WB	MCDBR	18.4	17.7	16.7	16.6	16.8	17.7	18.7	20.2	18.4	18.5	18.3	20.3
		MCWBR	14.0	12.6	10.7	9.7	7.3	5.8	5.9	5.9	6.2	9.4	11.6	16.9
(40) Clear-Sky Solar Irradiance	taub		0.330	0.347	0.368	0.419	0.469	0.448	0.469	0.458	0.428	0.377	0.370	0.329
	taud		2.494	2.459	2.400	2.266	2.170	2.281	2.251	2.277	2.319	2.442	2.399	2.507
	Ebn at Noon		286	291	291	278	264	267	261	264	268	277	268	281
	Edh at Noon		29	32	36	43	48	43	44	42	39	32	31	27
(44) All-Sky Solar Radiation	RadAvg		915	1034	1395	1690	1841	2013	2030	1919	1619	1356	1012	808
	RadStd		91	194	128	127	137	149	160	113	151	171	111	101

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+157

Nomenclature: See separate page